

SUPPLEMENT TO NEURAL BELLMAN OPERATORS: REVISION R29

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1. REVIEW OBJECT AND PRESERVATION

The article, this supplement, the computational appendix, the point-by-point response, and the historical annex form one revision object. The base evidence commit is `cd1191f278948942f42c6507b4566108f2aa2f6d`; the latest addressed report is R27, dated September 24, 2026, at `d6f6f7fa2b0ef40e42206dc64406f004cc83c71e`. R26 and the independent R25 second pass are addressed individually in the response. No earlier source, result, or review branch is overwritten. The old root index is retained in the source audit. The original R27 raw-run files were absent from its reviewed tree; a later re-execution cannot supply an earlier freeze event.

2. EXACT ARITHMETIC AND INDEPENDENT CHECKS

The inventory implementation uses arbitrary-precision Python integers, NumPy object arrays, and exact Fraction comparisons. At (d, L) , rewards have denominator $100d$. Backward denominator recursion multiplies by $100D$ per stage, where $D = d + 2$ is the number of equiprobable demand outcomes. Hence action deficits, envelope comparisons, and tolerance acceptance use integers after exact scaling. Printed decimals are explanatory only; bounds are rounded outward by the table renderer. The independently implemented scalar reference enumerates primitive transitions and rewards, rather than calling the vectorized action-value or completion routines.

For each model, candidate and completion files are sealed before reference creation. The recorded dependency audit prohibits reference files and the optimal solver during generation. This establishes the program’s information path, not blinding of the investigator to earlier results. Each policy’s feasibility, settlement convention, own-value recursion, residual envelope, and every reference gap are checked. Uniform and budgeted completion both improve the raw policy pointwise on this graph. Hash and event-order checks are supplemented by unit tests of infeasible actions, settlement, exact boundary comparisons, and forbidden reference reads.

3. CONTROLS AND CORRECTION BURDEN

A myopic action maximizes the exact one-period reward. The base-stock policy orders toward $\lceil (L - 1)/2 \rceil$, choosing the lowest-index most deficient good and at most two units. The two-period comparator uses the exact terminal payoff at the endpoint of a truncated two-step horizon; it does not read the full-horizon reference. Random policies are sampled independently from feasible actions at each nonsettled state and time. All receive the same repair rules and budgets. The fixed cohort, including the pre-execution seed-index clarification, is preserved in the protocol files.

TABLE I: Proposal quality and median correction burden; percentages exclude settled states

d/L	Proposal	Cases	Raw regret max	Fixed %	Budget %	Repair s
2/4	neural	8	0	0	0	0.001778
2/4	polynomial	2	0.1391	5.833	5.833	0.001871
2/4	random	10	5.054	63.33	63.33	0.002297
2/4	myopic	1	3.849	81.67	81.67	0.002414
2/4	base_stock	1	1.935	48.33	48.33	0.002281
2/4	tabular2	1	0.8426	26.67	26.67	0.002075
3/4	neural	8	0.01418	0.2976	0.2976	0.006931
3/4	polynomial	2	0	0	0	0.006899
3/4	random	10	5.141	69.25	69.15	0.009537
3/4	myopic	1	3.59	87.9	87.9	0.009774
3/4	base_stock	1	1.339	39.48	39.48	0.008217
3/4	tabular2	1	0.7878	18.45	18.45	0.00757
4/4	neural	16	0.3666	2.868	2.255	0.03275
4/4	polynomial	2	0.3666	1.863	1.225	0.03208
4/4	random	10	5.736	73.43	73.21	0.04108
4/4	myopic	1	3.273	86.42	86.42	0.04022
4/4	base_stock	1	1.245	36.13	35.54	0.03699
4/4	tabular2	1	0.8514	17.5	17.45	0.03238
4/5	neural	5	0.3664	2.244	1.643	0.08003
4/5	polynomial	2	0.1925	1.763	1.102	0.08268
4/5	random	10	6.933	77.9	77.7	0.1064
4/5	myopic	1	4.057	83.79	83.79	0.1005
4/5	base_stock	1	1.884	39.28	39.04	0.09759
4/5	tabular2	1	1.491	25.56	25.56	0.08222
5/4	neural	1	0.2399	0.2933	0.2933	0.1362
5/4	polynomial	1	0.2399	0.4643	0.4032	0.1456
5/4	random	1	7.066	75.77	75.6	0.1838
5/4	myopic	1	3.552	86.46	86.46	0.1783
5/4	base_stock	1	2.544	30.36	30.36	0.1526
5/4	tabular2	1	2.72	17.69	17.69	0.1471
6/4	neural	1	0.8042	0.8364	0.8364	0.6143
6/4	polynomial	1	0.865	1.004	1.004	0.6497
6/4	random	1	7.541	77.65	77.65	0.8053
6/4	myopic	1	3.543	86.88	86.88	0.8977
6/4	base_stock	1	4.752	25.4	25.4	0.7107
6/4	tabular2	1	5.346	14.04	14.04	0.6834
7/4	neural	1	1.549	0.4998	0.4837	3.356
7/4	polynomial	1	1.809	1.227	1.211	3.542
7/4	random	1	10.41	78.46	78.46	4.315
7/4	myopic	1	3.659	87.25	87.24	4.151
7/4	base_stock	1	7.072	22.16	22.14	3.812
7/4	tabular2	1	8.011	10.46	10.46	3.411

4. TOLERANCE SENSITIVITY

The same 14 policies are evaluated at all three tolerances; these repeated audits are not independent replications. Changes depend on the completed suffix, so the table is descriptive rather than a global minimum-edit theorem.

TABLE II: Identical raw candidates under fixed and state-dependent budgets

d/L	Policy	ε	Fixed changes	Budget changes	Fixed bound	Budget bound
2/4	N32/2/301	1/1000	0	0	0	0
2/4	N32/2/301	1/100	0	0	0	0
2/4	N32/2/301	1/20	0	0	0	0
2/4	Myopic	1/1000	98	98	0	0
2/4	Myopic	1/100	98	98	0	0
2/4	Myopic	1/20	98	98	0	0
3/4	N32/2/301	1/1000	1	1	0	0
3/4	N32/2/301	1/100	1	1	0	0
3/4	N32/2/301	1/20	1	0	0	0.01355
3/4	Myopic	1/1000	443	443	0	0
3/4	Myopic	1/100	443	443	0	0
3/4	Myopic	1/20	443	440	0	0.01355
4/4	Myopic	1/1000	1763	1763	0	0
4/4	Myopic	1/100	1763	1763	0	0
4/4	Myopic	1/20	1763	1759	0	0.0325
4/4	N32/2/303	1/1000	61	61	0	0
4/4	N32/2/303	1/100	61	45	0	0.009483
4/4	N32/2/303	1/20	60	32	0.002989	0.0458
4/5	Myopic	1/1000	4183	4183	0	0
4/5	Myopic	1/100	4183	4183	0	0
4/5	Myopic	1/20	4183	4111	0	0.03431
4/5	N32/2/303	1/1000	118	113	0	3.962×10^{-4}
4/5	N32/2/303	1/100	113	82	3.962×10^{-4}	0.009483
4/5	N32/2/303	1/20	95	35	0.006091	0.04628
5/4	N32/2/303	1/1000	24	24	0	0
5/4	N32/2/303	1/100	24	24	0	0
5/4	N32/2/303	1/20	24	13	0	0.04974
5/4	Myopic	1/1000	7076	7076	0	0
5/4	Myopic	1/100	7076	7076	0	0
5/4	Myopic	1/20	7076	7061	0	0.04995
6/4	N32/2/303	1/1000	274	274	0	0
6/4	N32/2/303	1/100	274	274	0	0
6/4	N32/2/303	1/20	274	231	0	0.04879
6/4	Myopic	1/1000	28468	28462	0	5.167×10^{-4}
6/4	Myopic	1/100	28462	28462	5.167×10^{-4}	5.167×10^{-4}
6/4	Myopic	1/20	28462	28456	5.167×10^{-4}	0.01098
7/4	N32/2/303	1/1000	655	655	0	0
7/4	N32/2/303	1/100	655	634	0	0.009704
7/4	N32/2/303	1/20	655	537	0	0.02474
7/4	Myopic	1/1000	114348	114348	0	0
7/4	Myopic	1/100	114348	114341	0	0.004821
7/4	Myopic	1/20	114341	114341	0.004821	0.004821

5. MEMORY, BIT GROWTH, AND ENUMERATION WORK

TABLE III: Storage and arithmetic coverage; RSS is a process high-water mark in KiB

d/L	Model B	Policy B	Feasible terms	Dense slots	Bits	RSS KiB
2/4	6112	1024	1632	2560	80	322444
3/4	40672	4096	11880	17920	84	323596

d/L	Model B	Policy B	Feasible terms	Dense slots	Bits	RSS KiB
4/4	244448	16384	73296	110592	86	343612
4/5	596105	40000	197568	270000	87	349856
5/4	1371856	65536	415128	630784	89	365856
6/4	7328752	262144	2227392	3407872	91	465000
7/4	37733696	1048576	11500488	17694720	93	968908

6. ORIGINAL-ECONOMY ORACLE AND PRODUCTION EVIDENCE

The article gives the complete likelihood-ratio proof, analytic remainder construction, projected-poll theorem, and transport identity. The following tables retain the executed precision levels and matched production comparison. Class regret is relative to the fixed scalar preference-adjustment class, not the original unrestricted policy problem. A successful pointwise interval gate is a payoff comparison, whereas a smaller original full-domain bound is only a certificate improvement. The original financed dual-construction time remains unmeasured; all recorded central timing totals state their conditional scope.

TABLE IV: All executed scalar polling meshes and precision-dependent work

k	h	θ	Calls	$\ G_1\ $ bound	Seconds	Call bound
1/2	1/25	1/5	17	0.04001	0.3453	257166
1/2	1/50	1/5	2	0.02001	0.07249	975450
1/2	1/100	1/5	2	0.01001	0.07256	3901800
1/2	1/200	1/5	2	0.005001	0.07264	15607200
2	1/25	1/5	17	0.06211	0.3399	257166
2	1/50	9/50	5	0.03106	0.1249	997506
2	1/100	9/50	3	0.01553	0.08933	3988965
2	1/200	37/200	6	0.007763	0.1434	15955851

TABLE V: Analytic derivative remainder budgets (orders zero, one, and two)

k	Order	Flow	Terminal	Stopping	Total
1/2	0	1.51×10^{-15}	9.451×10^{-25}	6.198×10^{-30}	1.51×10^{-15}
1/2	1	3.019×10^{-14}	1.898×10^{-22}	2.673×10^{-13}	2.974×10^{-13}
1/2	2	1.208×10^{-12}	3.872×10^{-20}	7.56×10^{-12}	8.768×10^{-12}
2	0	1.51×10^{-15}	9.451×10^{-25}	6.198×10^{-30}	1.51×10^{-15}
2	1	3.019×10^{-14}	1.898×10^{-22}	2.673×10^{-13}	2.974×10^{-13}
2	2	1.208×10^{-12}	3.872×10^{-20}	7.566×10^{-12}	8.773×10^{-12}

TABLE VI: Full-path displacement and moment diagnostics

Seed	Order	Min step	Median	95th pct.	m discrepancy	v discrepancy
27101	8/16	0.0101	0.03226	3.328	1.148×10^{-18}	4.765×10^{-22}
27101	12/24	0.0101	0.03233	3.328	1.789×10^{-18}	3.706×10^{-22}
27102	8/16	0.01073	0.03079	2.493	2.386×10^{-18}	3.177×10^{-22}

Seed	Order	Min step	Median	95th pct.	m discrepancy	v discrepancy
27102	12/24	0.01073	0.03074	2.493	1.952×10^{-18}	5.294×10^{-22}
27103	8/16	0.02044	0.0489	3.154	7.387×10^{-18}	5.294×10^{-22}
27103	12/24	0.02038	0.04887	3.153	5.34×10^{-18}	4.236×10^{-22}

TABLE VII: Paired final-payoff interval: gated minus ungated

Seed	Method	Lower	Upper	Proved positive	Rejections
27201	Neural Adam	-3.9377×10^{-7}	3.9377×10^{-7}	no	0
27201	Direct Adam	-3.9377×10^{-7}	3.9377×10^{-7}	no	0
27201	Direct L-BFGS-B	-3.9397×10^{-7}	3.9357×10^{-7}	no	3
27202	Neural Adam	-3.9377×10^{-7}	3.9377×10^{-7}	no	0
27202	Direct Adam	-3.9377×10^{-7}	3.9377×10^{-7}	no	0
27202	Direct L-BFGS-B	-3.9365×10^{-7}	3.9389×10^{-7}	no	3

7. INTERPRETATION OF EXACT-COMMIT VALIDATION

The read-only validation workflow checks the final source commit from a clean checkout, verifies frozen scientific hashes, rebuilds the PDFs, and records the exact SHA, run identifier, and artifact identity in its output. The record is external to the commit it identifies, avoiding a self-referential hash assertion. Compilation, tests, and workflow success do not prove uninstantiated hypotheses of the original continuous-control theorems.